CS1231S Discrete Structures

WEEK 9 RECITATION
LECTURE 10. COUNTING AND PROBABILITY I

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CS1231S (AY2026/27 Sem1) QnA

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Applications of Counting in Computer Science

Concepts	CS applications	Typical question
Multiplication rule	Passwords, identifiers, state spaces	How many 8-character passwords are possible?
Addition rule	Alternative program cases/formats	How many valid inputs belong to one of several disjoint formats?
Permutations	Scheduling, ordering, search algorithms	In how many orders can n jobs be scheduled?
Combinations	Selecting features, nodes, edges, test cases	How many pairs of computers are there in a network of n computers?
Binomial coefficients	Algorithms, probability, Hamming distance	How many n-bit strings contain exactly k ones?
Inclusion-exclusion	Databases, search/filtering, counting constrained objects	How many users satisfy condition A or B without double-counting?
Probability + counting	Hashing, compression, collision arguments	What is the probability that two hash values collide?
Pigeonhole principle	Hashing, compression, collision arguments	Why must collisions occur if more keys than hash slots exist?

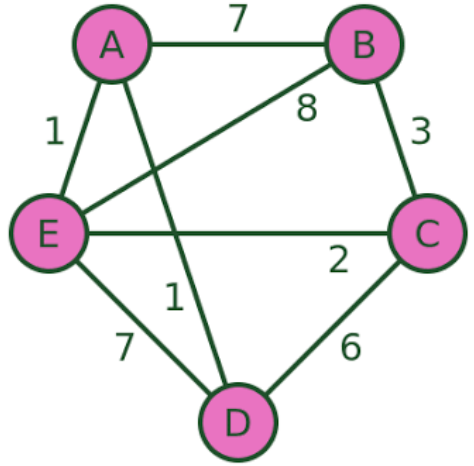
Applications of Counting in Computer Science

Eg: Algorithm analysis

Suppose an algorithm tries every subset of n items.
The number of possibilities is $\sum_{k=0}^{n} \binom{n}{k} = 2^n$

Immediately, counting tells us why brute force becomes infeasible.

For the Travelling Salesman Problem, fixing the starting city leaves roughly $(n-1)!$ possible tours to examine.
So permutations aren't merely a mathematical exercise – they explan combinatorial explosion and why we need clever algorithms, heuristics, dynamic programming, branch-and-bound, etc.



Two coins



- 0, 1 or 2 heads?
- Does each of these events occur about 1/3 of the time?

1/4



No heads obtained

2/4



One head obtained



1/4



Two heads obtained

Sample Space, Event, Equally Likely Probability

To say that a process is **random** means that when it takes place, one outcome from some set of outcomes is sure to occur, but it is impossible to predict with certainty which outcome that will be.

Definitions

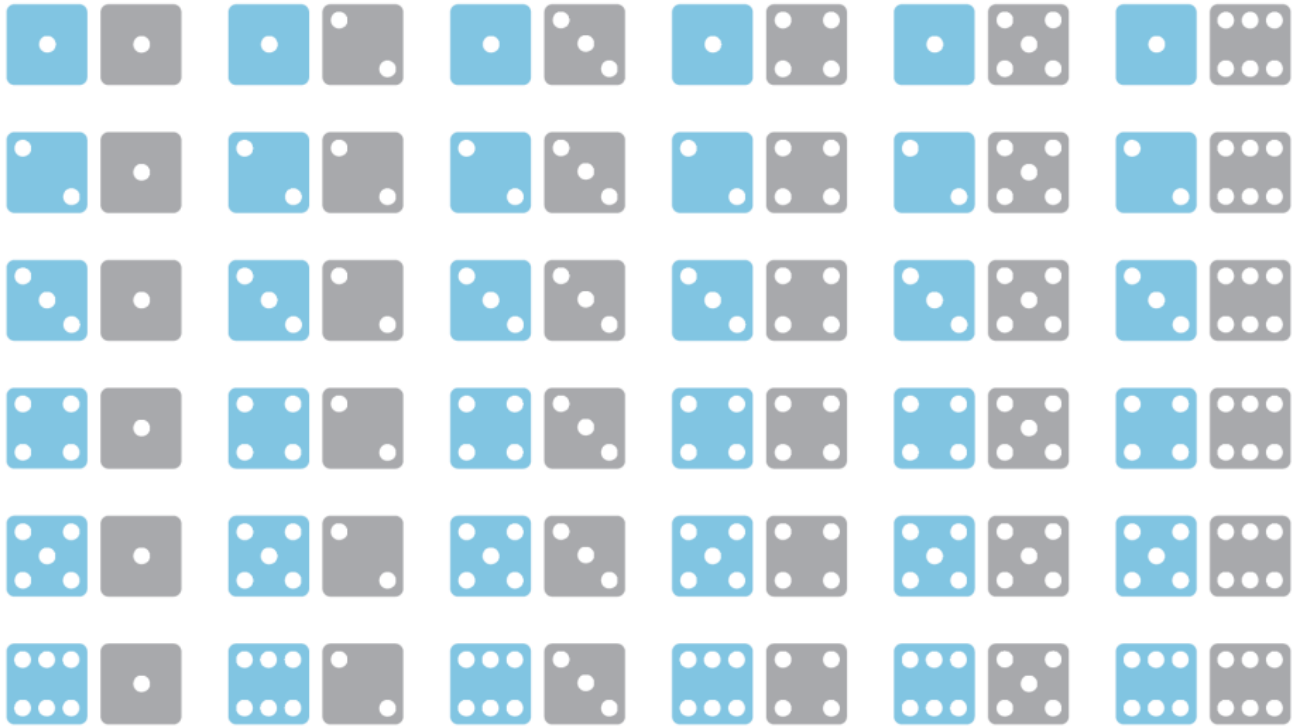
A **sample space** is the set of all possible outcomes of a random process or experiment. An **event** is a subset of a sample space.

Equally Likely Probability Formula

If S is a finite sample space in which all outcomes are equally likely and E is an event in S, then the **probability** of E, denoted $P(E)$, is

$$P(E) = \frac{\text{The number of outcomes in } E}{\text{The total number of outcomes in } S} = \frac{|E|}{|S|}$$

Rolling a Pair of Dice





A more compact notation identifies, say,  with the notation 24,  with 53, and so forth.

a. Use the compact notation to write the sample space S of possible outcomes.

S = { 11, 12, 13, 14, 15, 16, 21, 22, 23, 24, 25, 26, 31, 32, 33, 34, 35, 36, 41, 42, 43, 44, 45, 46, 51, 52, 53, 54, 55, 56, 61, 62, 63, 64, 65, 66 }

b. Use set notation to write the event E that the numbers showing face up have a sum of 6 and find the probability of this event.

E = { 15, 24, 33, 42, 51 }

$P(E) = 5/36$

Counting Rules (1)

Theorem 9.1.1 The Number of Elements in a List

If m and n are integers and $m \leq n$, then there are $n - m + 1$ integers from m to n inclusive.

Eg: How many integers are there from 5 through 12?

$12 - 5 + 1 = 8$

a. How many 3-digit integers (from 100 to 999 inclusive) are divisible by 5?

100 101 102 103 104 **105** 106 107 108 109 **110** … 994 **995** 996 997 998 999

$5\times$**20** $5\times$**21** $5\times$**22** … $5\times$**199**

From Theorem 9.1.1, there are $199 - 20 + 1 =$ **180** such integers.

b. What is the probability that a randomly chosen 3-digit integer is divisible by 5?

By Theorem 9.1.1, total number of integers from 100 through 999 = $999 - 100 + 1 = 900$.

By part (a), 180 of these are divisible by 5.

Hence, answer = 180/900 = **1/5**.

Possibility Trees

Example 1: Teams *A* and *B* are to play each other repeatedly until one wins two games in a row, or a total of three games.

One way in which this tournament can be played is for *A* to win the first game, *B* to win the second, and *A* to win the third and fourth games. Denote this by writing *A*–*B*–*A*–*A*.

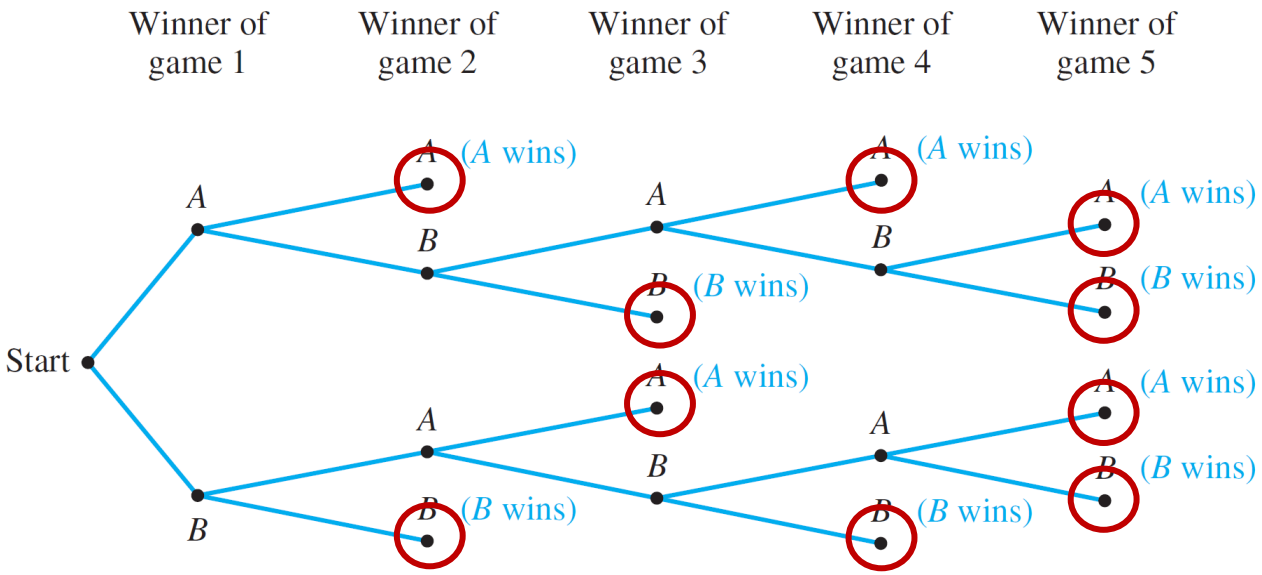


Figure 9.2.1 The Outcomes of a Tournament

a. How many ways can the tournament be played? 10

b. Assuming that all the ways of playing the tournament are equally likely, what is the probability that five games are needed to determine the tournament winner?

Probability that 5 games are needed = 4/10 = **2/5**

Counting Rules (2)

Theorem 9.2.1 The Multiplication/Product Rule

If an operation consists of k steps and

the first step can be performed in n_1 ways,

the second step can be performed in n_2 ways
(regardless of how the first step was performed),

:

the kth step can be performed in n_k ways
(regardless of how the preceding steps were performed),

Then the entire operation can be performed in

$$n_1 \times n_2 \times n_3 \times \ldots \times n_k \text{ ways.}$$

Principle of Product (or the Multiplication Principle)

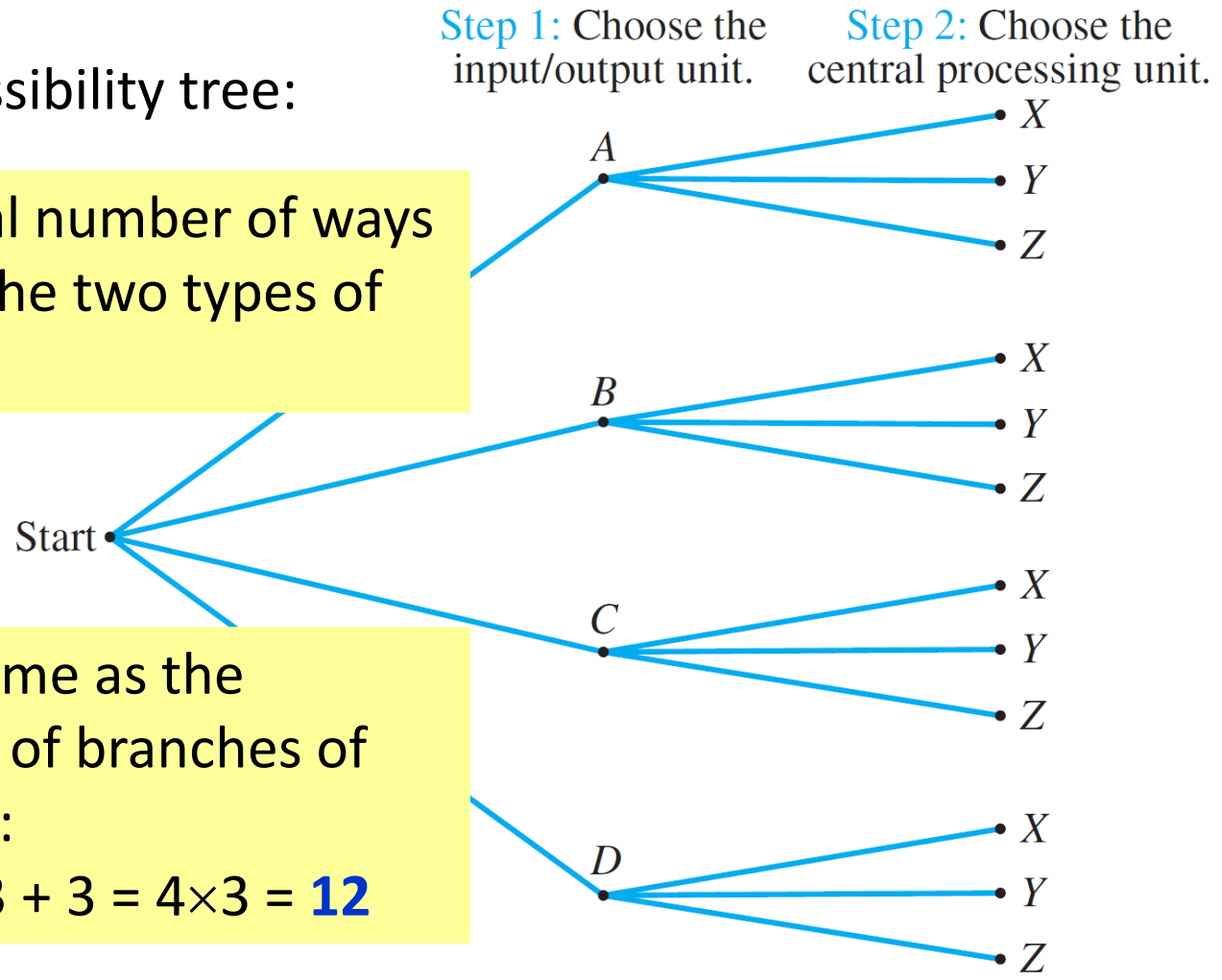
If there are m ways of doing something and n ways of doing another thing, then there are mn ways of performing both actions.

The Multiplication/Product Rule

Suppose a computer installation has four input/output units (*A*, *B*, *C*, and *D*) and three central processing units (*X*, *Y*, and *Z*).

Any input/output unit can be paired with any central processing unit. How many ways are there to pair an input/output unit with a central processing unit?

Possibility tree:



The total number of ways to pair the two types of units…

is the same as the number of branches of the tree:

3 + 3 + 3 + 3 = 4×3 = **12**

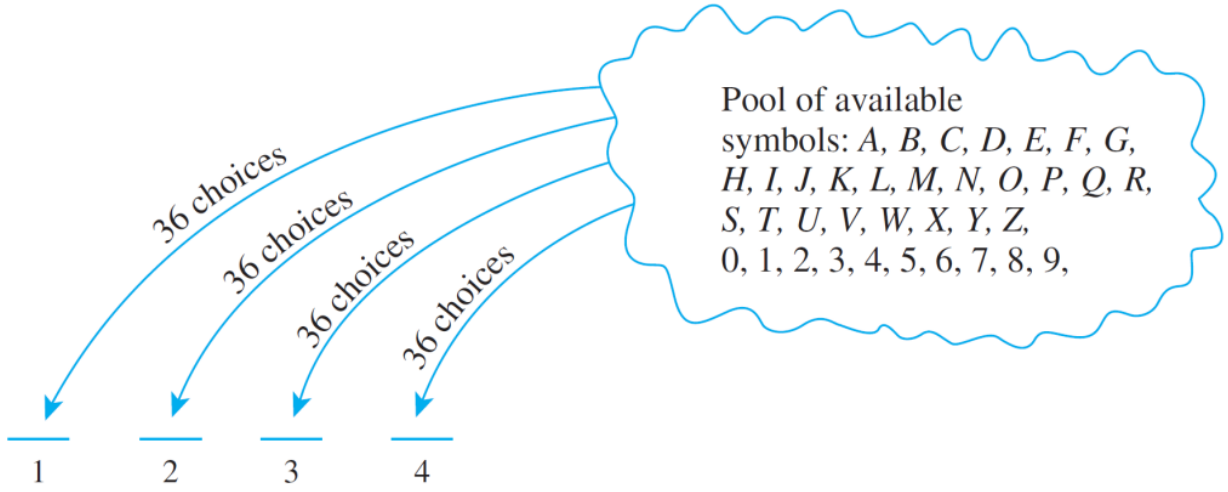
Figure 9.2.2 Pairing Objects Using a Possibility Tree

The Multiplication/Product Rule

Example 2: A typical PIN is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed. Examples: CARE, 3387, B32B, and so forth.

How many different PINs are possible?

You can think of forming a PIN as a four-step operation to fill in each of the four symbols in sequence.



Hence, by the **multiplication rule**, there are:
$36 \times 36 \times 36 \times 36 = 36^4 =$ **1,679,616** PINs in all.

The Multiplication/Product Rule

Example 3: A typical PIN is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed. Now, suppose that repetition is not allowed.

a. How many different PINs are there?

Step 1: Choose the 1st symbol.	36 ways	Hence, by the **multiplication rule**, there are:
Step 2: Choose the 2nd symbol.	35 ways	$36 \times 35 \times 34 \times 33 =$ **1,413,720** PINs in all with no repeated symbol.
Step 3: Choose the 3rd symbol.	34 ways	
Step 4: Choose the 4th symbol.	33 ways	

b. If all PINs are equally likely, what is the probability that a PIN chosen at random contains no repeated symbols?

1,679,616 PINS in all.
1,413,720 PINs with no repeated symbol.

Hence, probability that a PIN chosen at random contains no repeated symbols:

$$\frac{1,413,720}{1,679,616} \cong \mathbf{0.8417}$$

Power Set

Theorem 5.2.4 (Sets)

Suppose A is a finite set. Then $|\mathcal{P}(A)| = 2^{|A|}$.

Let $A = \{a_1, a_2, \cdots, a_n\}$. Then $|A| = n$.

A subset of A is formed by picking or dropping a_i from A, for $i = 1,2, \ldots, n$.

Examples:

$$\{\ a_1 \quad a_2 \quad a_3 \quad a_4 \quad \cdots \quad a_n\ \} \Rightarrow \{a_1, a_3, a_n\}$$
$$\checkmark \quad \times \quad \checkmark \quad \times \quad \times \quad \checkmark$$

$$\{\ a_1 \quad a_2 \quad a_3 \quad a_4 \quad \cdots \quad a_n\ \} \Rightarrow \emptyset$$
$$\times \quad \times \quad \times \quad \times \quad \times \quad \times$$

Since there are 2 choices (pick or drop) for every element a_i and there are n elements, by the multiplication rule there are 2^n ways of forming a subset.

When the Multiplication Rule is Difficult to Apply

Example 4: Consider the following problem:

Three officers – a president, a treasurer, and a secretary – are to be chosen from among four people: Ann, Bob, Cyd, and Dan. Suppose that, for various reasons, Ann cannot be president and either Cyd or Dan must be secretary. How many ways can the officers be chosen?

It is natural to try to solve this problem using the multiplication rule. A person might answer as follows:

There are three choices for president (all except Ann), three choices for treasurer (all except the one chosen as president), and two choices for secretary (Cyd or Dan).

Therefore, by the **multiplication rule**, $3 \times 3 \times 2 = \mathbf{18}$

Is this correct?

It is incorrect. The number of ways to choose the secretary varies depending on who is chosen for president and treasurer.

For instance, if Bob is chosen for president and Ann for treasurer, then there are two choices for secretary: Cyd and Dan.

But if Bob is chosen for president and Cyd for treasurer, then there is just one choice for secretary: Dan.

When the Multiplication Rule is Difficult to Apply

The clearest way to see all the possible choices is to construct the possibility tree, as shown below.

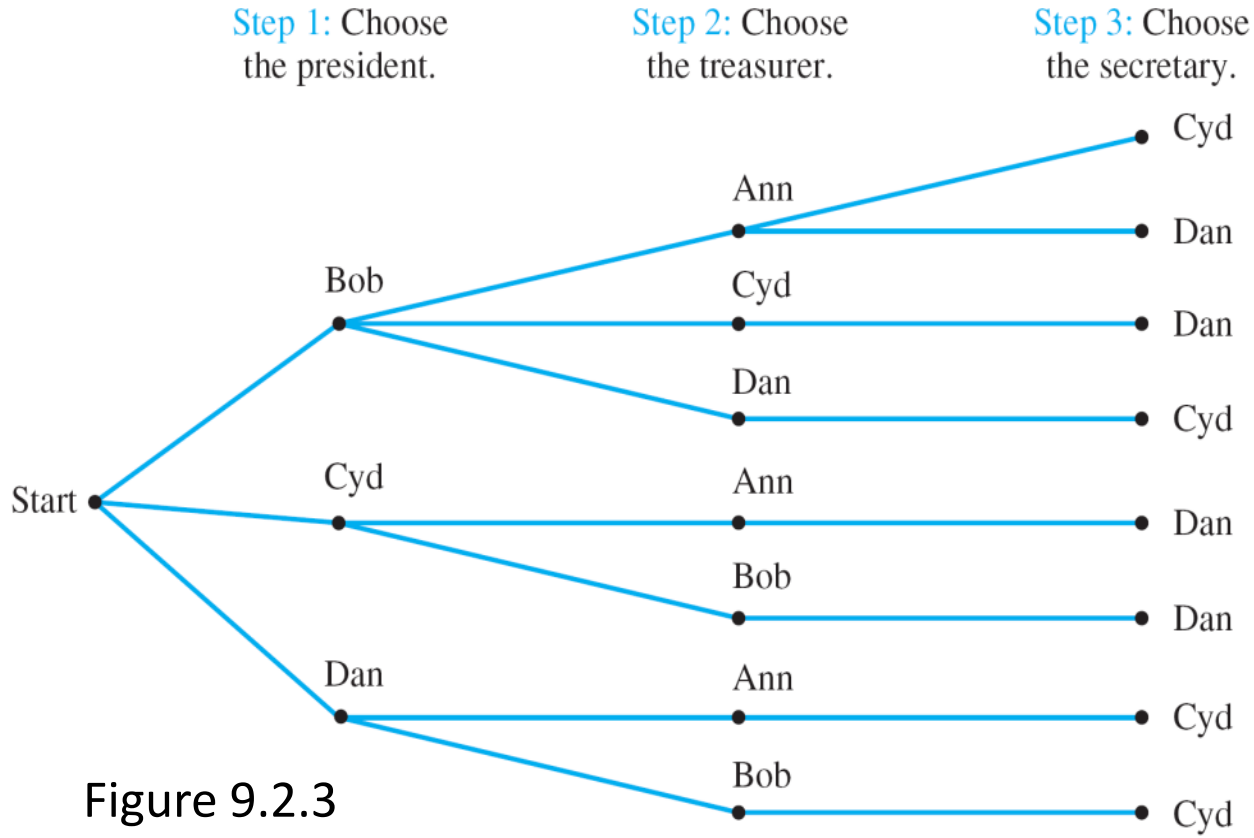


Figure 9.2.3

How many ways?

8 ways.

When the Multiplication Rule is Difficult to Apply

Example 5: Reorder the steps for choosing the officers in the previous example so that the total number of ways to choose officers can be computed using the multiplication rule.

Step 1: Choose the secretary.
Step 2: Choose the president.
Step 3: Choose the treasurer.

2 ways: Either Cyd or Dan.

2 ways: Neither Ann nor the person chosen in step 1 may be chosen, but either of the other two may.

2 ways: Either of the 2 persons not chosen in steps 1 and 2 may be chosen.

Hence, total number of ways = $2 \times 2 \times 2 = \mathbf{8}$

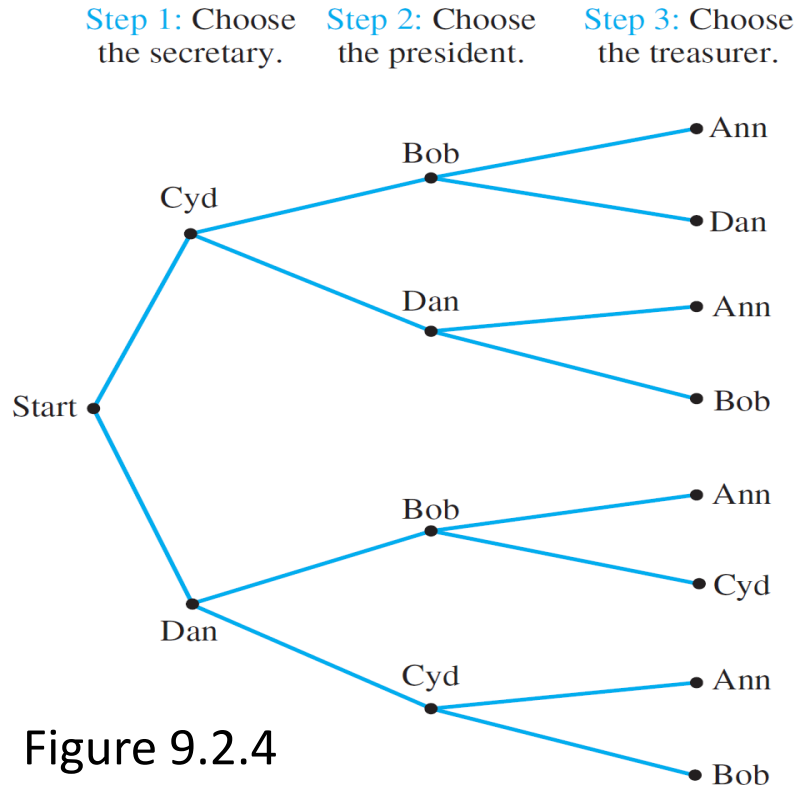


Figure 9.2.4

Counting Rules (3)

Theorem 9.3.1 The Addition/Sum Rule

Suppose a finite set A equals the union of k distinct mutually disjoint subsets $A_1, A_2, \ldots, A_k$. Then

$$|A| = |A_1| + |A_2| + \ldots + |A_k|.$$

Principle of Sum (or the Addition Principle)

If we have m ways of doing something and n ways of doing another thing and we cannot do both at the same time, then there are $m+n$ ways to choose one of these actions.

The **addition rule** (or **sum rule**) states that the number of elements in a union of mutually disjoint finite sets equals the sum of the number of elements in each of the component sets.

The Addition/Sum Rule

Example 7: A computer access password consists of from one to three letters chosen from the 26 letters in the alphabet with repetitions allowed. How many different passwords are possible?

The set of all passwords can be partitioned into subsets consisting of those of length 1, length 2, and length 3:

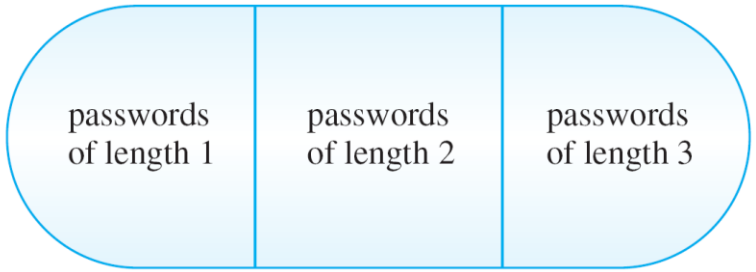


Figure 9.3.1 Set of all passwords of length ≤ 3

By the **addition rule**, the total number of passwords equals the sum of the number of passwords of length 1, length 2, and length 3.

Number of passwords of length 1 = 26

Number of passwords of length 2 = 26^2

Number of passwords of length 3 = 26^3

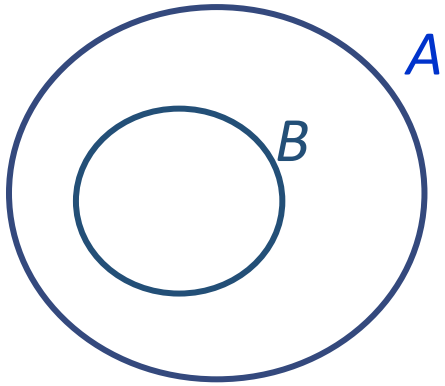
Hence, total number of passwords
= $26 + 26^2 + 26^3$ = **18,278**.

Counting Rules (4)

Theorem 9.3.2 The Difference Rule

If A is a finite set and $B \subseteq A$, then

$$|A \setminus B| = |A| - |B|.$$



The difference rule holds for the following reason:
If $B \subseteq A$, then B and $A \setminus B$ are disjoint and $B \cup (A \setminus B) = A$.

Hence, by the addition rule,

$$|B| + |A \setminus B| = |A|.$$

Subtracting $|B|$ from both sides gives the equation

$$|A \setminus B| = |A| - |B|.$$

The Difference Rule

Example 8: A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed.

a. How many PINs contain repeated symbols?

Repetitions allowed: $36^4 = 1{,}679{,}616$ PINs.

Repetitions not allowed: $36 \times 35 \times 34 \times 33 = 1{,}413{,}720$ PINS.

By the **difference rule**, there are 1,679,616 – 1,413,720 = **265,896** PINS that contain at least one repeated symbol.

The Difference Rule

Example 8: A typical PIN (personal identification number) is a sequence of any four symbols chosen from the 26 letters in the alphabet and the ten digits, with repetition allowed.

b. If all PINs are equally likely, what is the probability that a randomly chosen PIN contains a repeated symbol?

The probability that a randomly chosen PIN contains a repeated symbol is

$$\frac{265{,}896}{1{,}679{,}616} \cong \mathbf{0.158}$$

Probability of the Complement of an Event

The probability of the complement of an event is obtained by subtracting the probability of the event from the number 1.

Formula for the Probability of the Complement of an Event

If S is a finite sample space and A is an event in S, then

$$P(\bar{A}) = 1 - P(A)$$

Counting Rules (5)

Theorem 9.3.3 The Inclusion/Exclusion Rule for 2 or 3 Sets

If A, B, and C are any finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

and

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

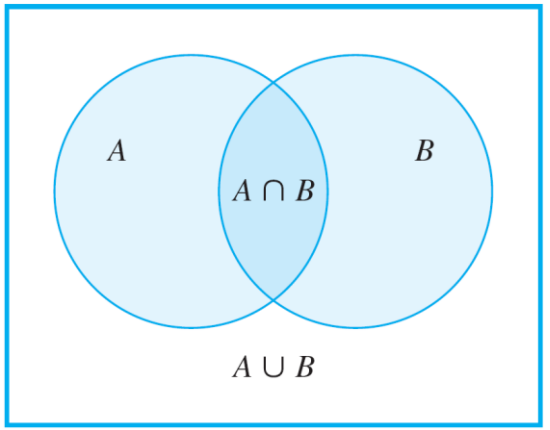
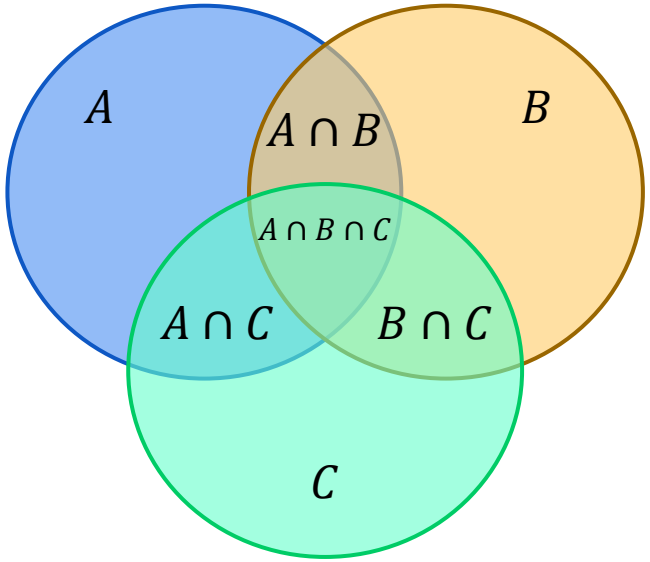
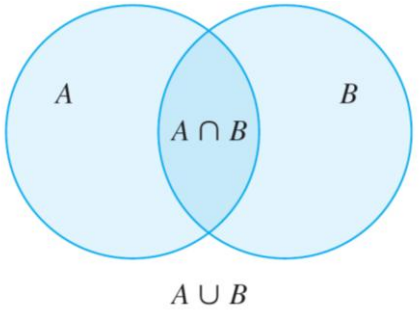


Figure 9.3.5



Inclusion-Exclusion Rule



Example 9: Counting elements of a general union.

a. How many integers from 1 through 1,000 are multiples of 3 or multiples of 5?

Let A = the set of all integers in [1, 1000] that are multiples of 3.
Let B = the set of all integers in [1, 1000] that are multiples of 5.

Then $A \cup B$ = the set of all integers in [1, 1000] that are multiples of 3 or 5.

Then $A \cap B$ = the set of all integers in [1, 1000] that are multiples of both 3 and 5
= the set of all integers in [1, 1000] that are multiples of 15.

Every multiple of 3 in [1, 1000] has the form $3k$, for some integer k in [1, 333].

Hence, $|A| = 333$

Every multiple of 5 in [1, 1000] has the form $5k$, for some integer k in [1, 200].

Hence, $|B| = 200$

Every multiple of 15 in [1, 1000] has the form $15k$, for some integer k in [1, 66].

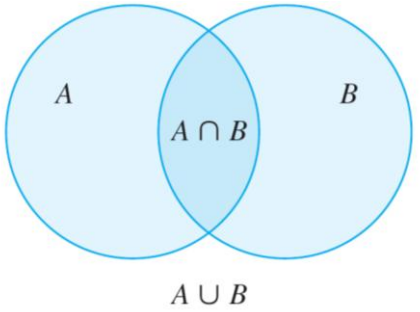
Hence, $|A \cap B| = 66$

By inclusion-exclusion rule,

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$= 333 + 200 - 66 = 467$$

Thus, **467** integers from 1 through 1000 are multiples of 3 or multiples of 5.

Inclusion-Exclusion Rule



Example 9: Counting elements of a general union.

b. How many integers from 1 through 1,000 are neither multiples of 3 nor multiples of 5?

There are 1000 integers in [1, 1000]. By part (a), 467 of these are multiples of 3 or multiples of 5.

Thus, by the **difference rule**, there are 1000 – 467 = **533** that are neither multiples of 3 nor multiples of 5.

Note that the solution to part (b) hid a use of De Morgan's law.

The number of elements that are neither in A nor in B is $|\bar{A} \cap \bar{B}|$.
By De Morgan's law, $\bar{A} \cap \bar{B} = (\overline{A \cup B})$.

So $|\overline{A \cup B}|$ was then calculated using the set difference rule:

$$|\overline{A \cup B}| = |U| - |A \cup B|$$

where the universe U was the set of all integers in [1, 1000].

Exercises



Exercise 1: A password consists of **6 characters**, each of which is either an uppercase letter or a digit. Repetition is allowed. How many passwords at least one letter and at least one digit?

Total passwords: 36^6

Passwords with only letters: 26^6

Passwords with only digits: 10^6

Answer: $36^6 - 26^6 - 10^6$

Exercises



Exercise 2: How many 4-digit even numbers are there?

(Note: 0123 is 3-digit and 0075 is 2-digit.)

Method 1:

1. The smallest 4-digit even number is 1000 and the largest is 9998.
2. The are 9000 (9999 – 1000 + 1) numbers in the range [1000,9999].
3. Half of them are even, hence 9000/2 = 4500.

Method 2:

1. 9 choices for thousands place.
2. 10 choices for hundreds place.
3. 10 choices for tens place.
4. 5 choices for ones place.
5. Therefore, 9 × 10 × 10 × 5 = 4500 (by multiplication rule)

Exercises



Exercise 3: A username consists of **3 lowercase letters followed by 3 digits.** Letters may be repeated, but digits may not. How many usernames satisfy both of the conditions below?

- At least one of the letters is a vowel;
- At least one of the digits is even.

Letters: $26^3 - 21^3$

Digits:

Ordered triples of distinct digits: $10 \cdot 9 \cdot 8 = 720$

Triples containing no even digit: $5 \cdot 4 \cdot 3 = 60$

Acceptable digit sequences: $720 - 60 = 660$

Answer: $(26^3 - 21^3) \cdot 660$

Exercises



Exercise 4: Let $\Sigma = \{a, b, c, d, e\}$. How many strings of length 8 over Σ contain each of a, b, c at least once?

Let A, B, C be the sets of strings missing a, b, c respectively.

$|A| = |B| = |C| = 4^8$

$|A \cap B| = |A \cap C| = |B \cap C| = 3^8$

$|A \cap B \cap C| = 2^8$

By inclusion-exclusion rule:

$|A \cup B \cup C| = 3(4^8) - 3(3^8) + 2^8$

Answer: $5^8 - (3(4^8) - 3(3^8) + 2^8)$

Exercises



Exercise 5: Let W_n be the number of ways to form a row of n children, where each child is either a boy or a girl, and no two boys are adjacent.

a. What is W_4?

$W_4 = 8$

~~BBBB~~	~~BGBB~~	~~GBBB~~	~~GGBB~~
~~BBBG~~	BGBG	~~GBBG~~	GGBG
~~BBGB~~	BGGB	GBGB	GGGB
~~BBGG~~	BGGG	GBGG	GGGG

b. Write the recurrence relation for W_n for $n \geq 1$.

$W_1 = 2$

$W_2 = 3$

$W_n = W_{n-1} + W_{n-2}$, for $n \geq 3$

Permutation

A **permutation** of a set of objects is an ordering of the objects in a row.
For example, the set of elements *a*, *b*, and *c* has six permutations.

abc *acb* *cba* *bac* *bca* *cab*

In general, given a set of *n* objects, how many permutations does the set have?

Imagine forming a permutation as an *n*-step operation:

Step 1: Choose an element to write first.	← *n* ways
Step 2: Choose an element to write second.	← *n* – 1 ways
Step 3: Choose an element to write third.	← *n* – 2 ways
:	
Step *n*: Choose an element to write *n*th.	← 1 way

By the **multiplication rule**, there are

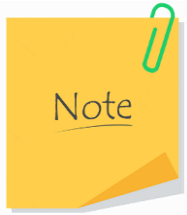
$n \times (n-1) \times (n-2) \times \ldots \times 2 \times 1 = \boldsymbol{n!}$

ways to perform the entire operation.

Permutation

Theorem 9.2.2 Permutations

The number of permutations of a set with n ($n \geq 1$) elements is $\boldsymbol{n!}$



0! is defined to be 1.

(See https://www.khanacademy.org/math/precalculus/x9e81a4f98389efdf:prob-comb/x9e81a4f98389efdf:combinatorics-precalc/v/zero-factorial-or-0)

Permutation

Example 6: Permutations of the letters in a word

a. How many ways can the letters in the word *COMPUTER* be arranged in a row?

All the eight letters in the word *COMPUTER* are distinct.
Hence, **8!** = **40320**.

b. How many ways can the letters in the word *COMPUTER* be arranged if the letters *CO* must remain next to each other (in order) as a unit?

7! = **5040**.

c. If letters of the word *COMPUTER* are randomly arranged in a row, what is the probability that the letters *CO* remain next to each other (in order) as a unit?

$$\frac{7!}{8!} = \frac{1}{8}$$

Permutation of Selected Elements

Given the set {*a*, *b*, *c*}, there are six ways to select two letters from the set and write them in order.

ab *ac* *ba* *bc* *ca* *cb*

Each such ordering of two elements of {*a*, *b*, *c*} is called a ***2-permutation*** of {*a*, *b*, *c*}?.

Definition

An ***r*-permutation** of a set of ***n* elements** is an ordered selection of *r* elements taken from the set.
The number of *r*-permutations of a set of *n* elements is denoted ***P*(*n*, *r*)**.

Other notations: ${}_nP_r$, P_r^n

Permutation of Selected Elements

Theorem 9.2.3 *r*-permutations from a set of *n* elements

If n and r are integers and $1 \leq r \leq n$, then the number of r-permutations of a set of n elements is given by the formula

$$P(n, r) = n(n-1)(n-2)\ldots(n-r+1)$$ first version

or, equivalently,

$$P(n, r) = \frac{n!}{(n-r)!}$$ second version

Permutation of Selected Elements

a. Evaluate $P(5, 2)$.

$P(5, 2) = 5! / (5 - 2)! = 5 \times 4 = \mathbf{20}$

b. How many 4-permutations are there of a set of seven objects?

$P(7, 4) = 7! / (7 - 4)! = 7 \times 6 \times 5 \times 4 = \mathbf{840}$

c. How many 5-permutations are there of a set of five objects?

$P(5, 5) = 5! / (5 - 5)! = 5! = \mathbf{120}$

Exercises



Eight people *A,B,C,D,E,F,G,H* are seated in a row.

Exercise 6: In how many arrangements are *A* and *B* seated next to each other?

Answer: $7!\times 2! = 10{,}080$

Exercise 7: In how many arrangements are *A* and *B* not seated next to each other?

Answer: $8! - 10{,}080 = 30{,}240$

Exercise 8: In how many arrangements are *A* and *B* next to each other, and *C* and *D* also next to each other?

Answer: $6!\times 2!\times 2! = 2880$

Exercises



Exercise 9: Seven students are arranged in a row for a photograph. Alice refuses to stand at either end of the row. How many arrangements are possible?

Answer: $5 \times 6! = 3600$

Alternatively: $7! - 2(6!) = 5040 - 1440 = 3600$

Exercise 10: Five boys and five girls are seated in a row. In how many arrangements do boys and girls alternate?

Two possible patterns: *BGBGBGBGBG* or *GBGBGBGBGB*

For each pattern, we can arrange the boys in $5!$ ways and girls in $5!$ ways.

Answer: $2(5!)(5!) = 28{,}800$

Exercises



Exercise 11: Five students *A,B,C,D,E* assigned five numbered seats. How many arrangements are there in which neither *A* nor *B* sits in their originally assigned seat?

Let *X* be the event "*A* sits in *A*'s assigned seat" and *Y* be the event "*B* sits in *B*'s assigned seat".

By inclusion-exclusion,

$$|X \cup Y| = |X| + |Y| - |X \cap Y| = 4! + 4! - 3! = 42.$$

Answer: $5! - 42 = 78$

The Pigeonhole Principle (PHP)



What injections and surjections tell us about cardinality

Pigeonhole Principle

Let A and B be finite sets. If there is an injection $f: A \to B$, then $|A| \leq |B|$.

Contrapositive: Let $m, n \in \mathbb{Z}^+$ with $m > n$. If m pigeons are put into n pigeonholes, then there must be (at least) one pigeonhole with (at least) two pigeons.

Dual Pigeonhole Principle

Let A and B be finite sets. If there is a surjection $f: A \to B$, then $|A| \geq |B|$.

Contrapositive: Let $m, n \in \mathbb{Z}^+$ with $m < n$. If m pigeons are put into n pigeonholes, then there must be (at least) one pigeonhole with no pigeons.

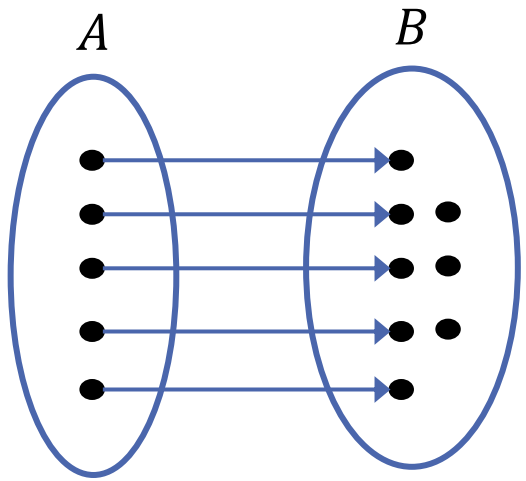
What injections and surjections tell us about cardinality

Pigeonhole Principle

Let A and B be finite sets. If there is an injection $f: A \to B$, then $|A| \leq |B|$.

Proof

1. Note that A is finite. Suppose $A = \{a_1, a_2, \cdots, a_m\}$ where $m = |A|$.
2. The injectivity of f tells us that, if $a_i \neq a_j$, then $f(a_i) \neq f(a_j)$.
3. So $f(a_1), f(a_2), \cdots, f(a_m)$ are m different elements of B.
4. This shows $|B| \geq m = |A|$.



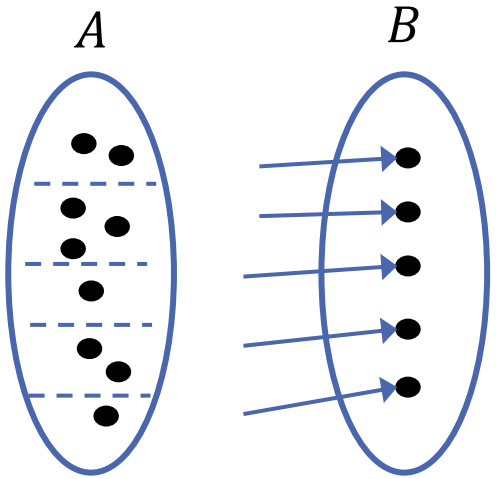
What injections and surjections tell us about cardinality

Dual Pigeonhole Principle

Let A and B be finite sets. If there is a surjection $f: A \to B$, then $|A| \geq |B|$.

Proof

1. Note that B is finite. Suppose $B = \{b_1, b_2, \cdots, b_n\}$ where $n = |B|$.
2. For each b_i, use the surjectivity of f to find $a_i \in A$ such that $f(a_i) = b_i$.
3. If $b_i \neq b_j$, then $f(a_i) \neq f(a_j)$ and so $a_i \neq a_j$ as f is a function.
4. So $a_1, a_2, \cdots, a_n$ are n different elements of A.
5. This shows $|A| \geq n = |B|$.



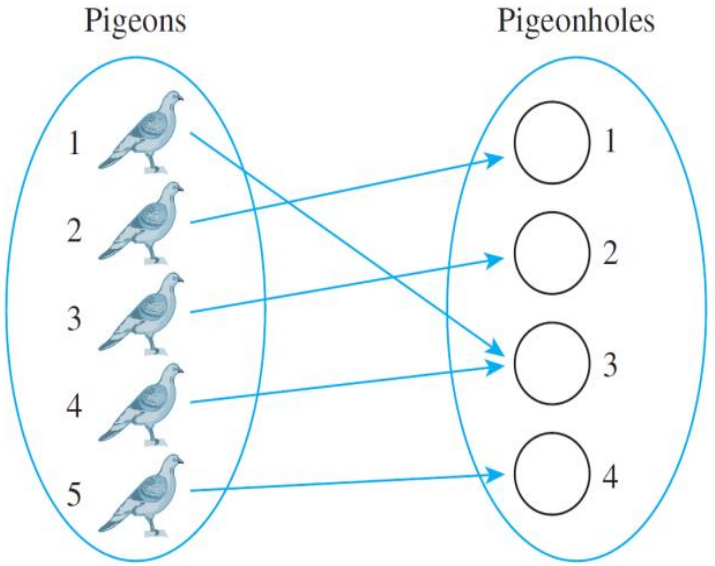
The Pigeonhole Principle (PHP)

The pigeonhole principle is sometimes called the *Dirichlet box principle* because it was first stated formally by J. P. G. L. Dirichlet (1805–1859).

Mathematical formulation:

Pigeonhole Principle (PHP)

A function from one finite set to a smaller finite set cannot be one-to-one: There must be at least 2 elements in the domain that have the same image in the co-domain.



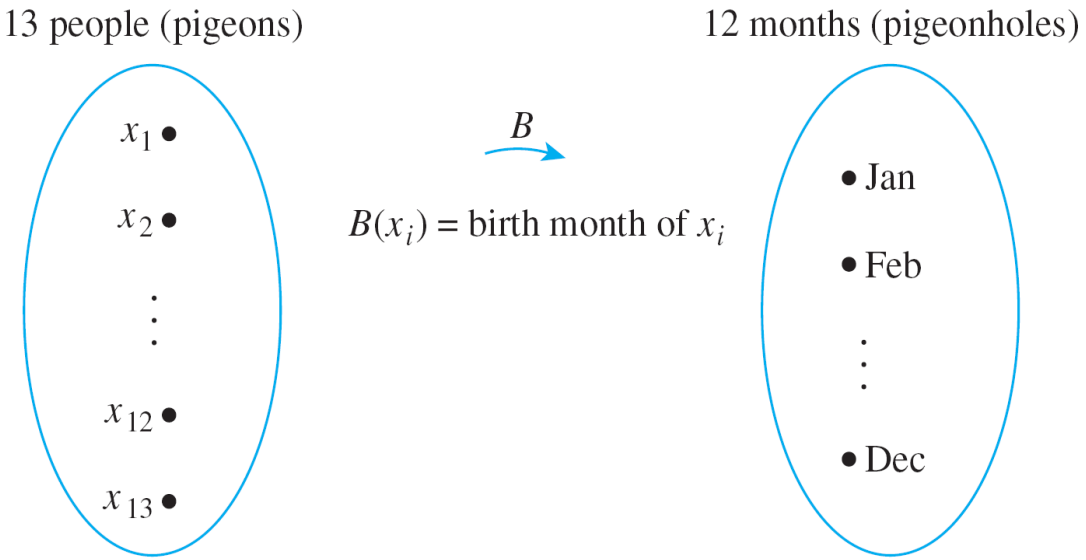
The Pigeonhole Principle (PHP)



Example 10:

a. In a group of six people, must there be at least two who were born in the same month? No.

In a group of 13 people, must there be at least two who were born in the same month? Why?



Yes. At least 2 people must have been born in the same month.

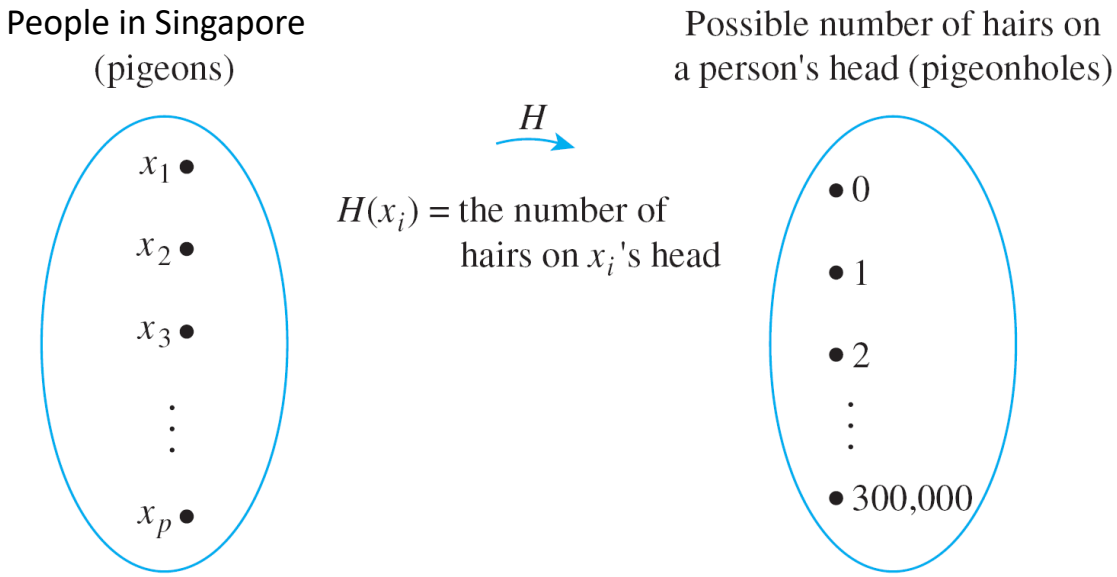
The Pigeonhole Principle (PHP)



Example 10:

b. Among the population of Singapore, are there at least two people with the same number of hairs on their heads? Why?

Population of Singapore: 6.21m (June 2026).
Hairs on head: average up to 150,000; no more than 300,000.



Yes. At least 2 people must have the same number of hairs on their heads.

Decimal Expansions of Fractions

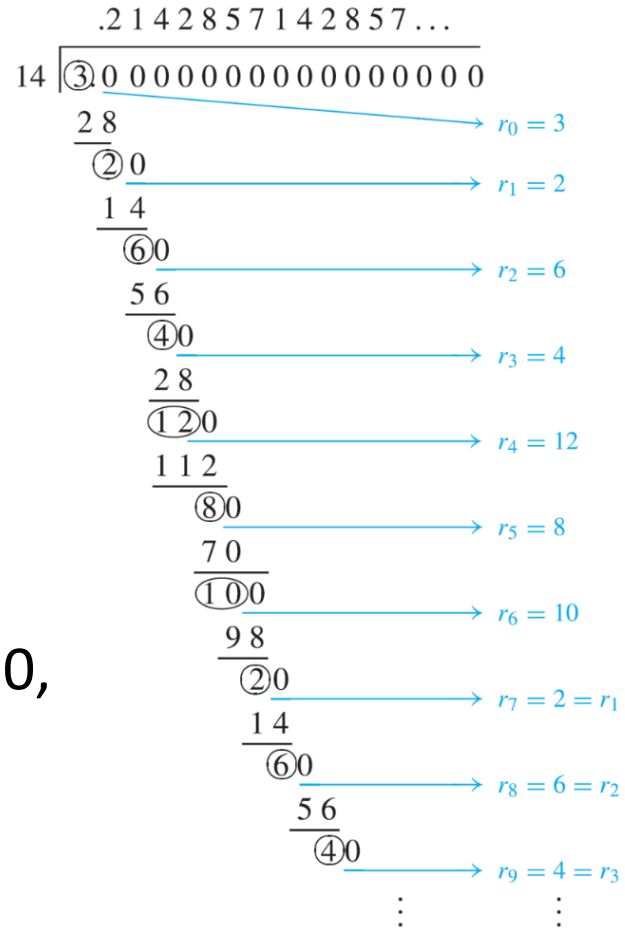
The decimal expansion of any rational number either terminates or repeats.

The decimal expansion of a/b is obtained by dividing a by b as illustrated here for $a = 3$ and $b = 14$.

Let $r_0 = a$ and let $r_1, r_2, r_3, \ldots$ be the successive remainders obtained in the long division of a by b.

By the quotient-remainder theorem (Theorem 8.1.16: Division Theorem), each remainder must be between 0 and $b - 1$. (Here, b is 14, and so the remainders are from 0 to 13.)

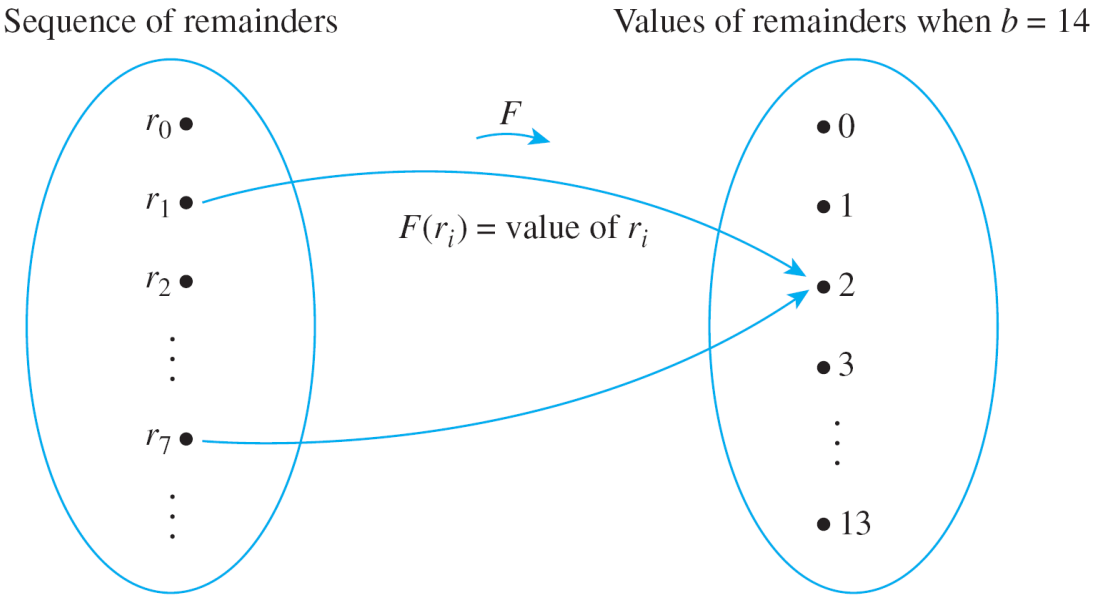
If some remainder $r_i = 0$, then the division terminates and a/b has a terminating decimal expansion. If no $r_i = 0$, then the division process and hence the sequence of remainders continues forever.



Decimal Expansions of Fractions

By the PHP, since there are more remainders than values that the remainders can take, some remainder value must repeat: $r_j = r_k$, for some indices j and k with $j < k$.

It follows that the decimal digits obtained from the divisions between r_j and r_{k-1} repeat forever.



In the case of 3/14, the repetition begins with $r_7 = 2 = r_1$ and the decimal expansion repeats the quotients obtained from the divisions from r_1 through r_6 forever:

$$3/14 = 0.2\overline{142857}$$

Generalized Pigeonhole Principle

If n pigeons fly into m pigeonholes and, for some positive integer k, $k < n/m$, then at least one pigeonhole contains $k + 1$ or more pigeons.

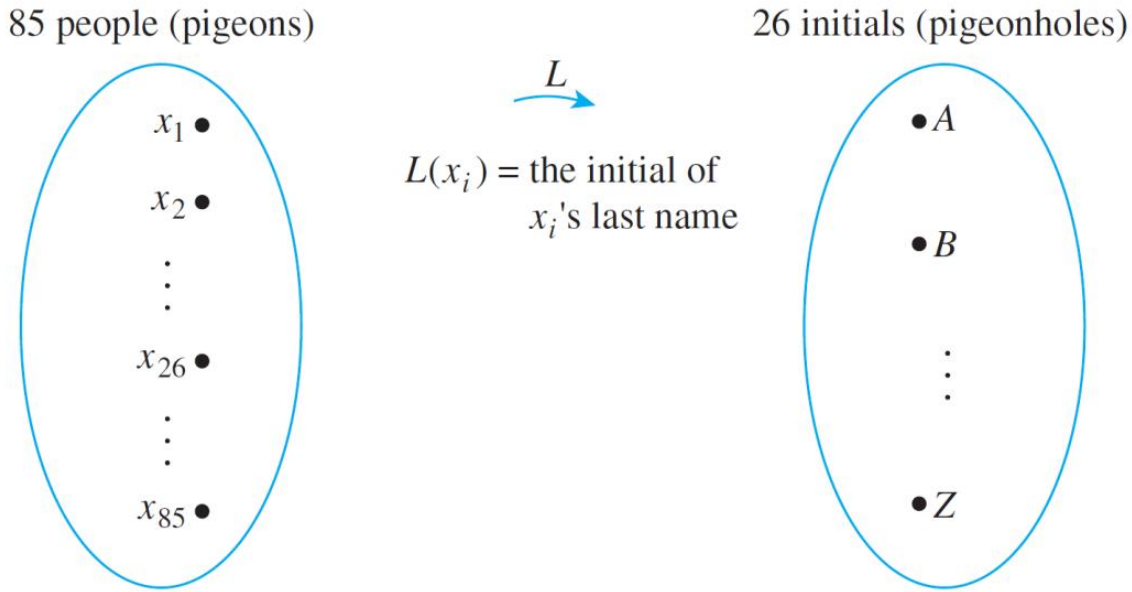
Generalized Pigeonhole Principle

For any function f from a finite set X with n elements to a finite set Y with m elements and for any positive integer k, if $k < n/m$, then there is some $y \in Y$ such that y is the image of at least $k + 1$ distinct elements of X.

Generalized Pigeonhole Principle

Example 11: Show how the generalized pigeonhole principle implies that in a group of 85 people, at least 4 must have the same last initial ('A', 'B', …, 'Z').

In this example the pigeons are the 85 people and the pigeonholes are the 26 possible last initials of their names. Note that $3 < 85/26 \cong 3.27$



Since $3 < 85/26$, the generalized pigeonhole principle states that some initial must be the image of at least four (3 + 1) people.

Thus, at least 4 people have the same last initial.

Generalized Pigeonhole Principle (contrapositive)

Generalized Pigeonhole Principle (Contrapositive Form)

For any function f from a finite set X with n elements to a finite set Y with m elements and for any positive integer k, if for each $y \in Y$, $f^{-1}(\{y\})$ has at most k elements, then X has at most km elements; in other words, $n \leq km$.

Generalized Pigeonhole Principle (contrapositive)

Example 11: Show how the generalized pigeonhole principle implies that in a group of 85 people, at least 4 must have the same last initial ('A', 'B', …, 'Z').

Generalized Pigeonhole Principle (Contrapositive Form)

For any function f from a finite set X with n elements to a finite set Y with m elements and for any positive integer k, if for each $y \in Y$, $f^{-1}(\{y\})$ has at most k elements, then X has at most km elements; in other words, $n \leq km$.

Suppose no 4 people out of the 85 had the same last initial. Then at most 3 would share any particular one.

By the generalized pigeonhole principle (contrapositive form), this would imply that the total number of people is at most $3 \times 26 = 78$. But this contradicts the fact that there are 85 people in all.

Hence at least 4 people share a last initial.

Generalized Pigeonhole Principle (contrapositive)

Example 12: There are 42 students who are to share 12 computers. Each student uses exactly 1 computer, and no computer is used by more than 6 students. Show that at least 5 computers are used by 3 or more students.

Using an Argument by Contradiction: Suppose not. Suppose that 4 or fewer computers are used by 3 or more students. *[A contradiction will be derived.]* Then 8 or more computers are used by 2 or fewer students.

Divide the set of computers into two subsets: C_1 and C_2.

Generalized Pigeonhole Principle (contrapositive)

Into C_1 place 8 of the computers used by 2 or fewer students; into C_2 place the computers used by 3 or more students plus any remaining computers (to make a total of 4 computers in C_2).

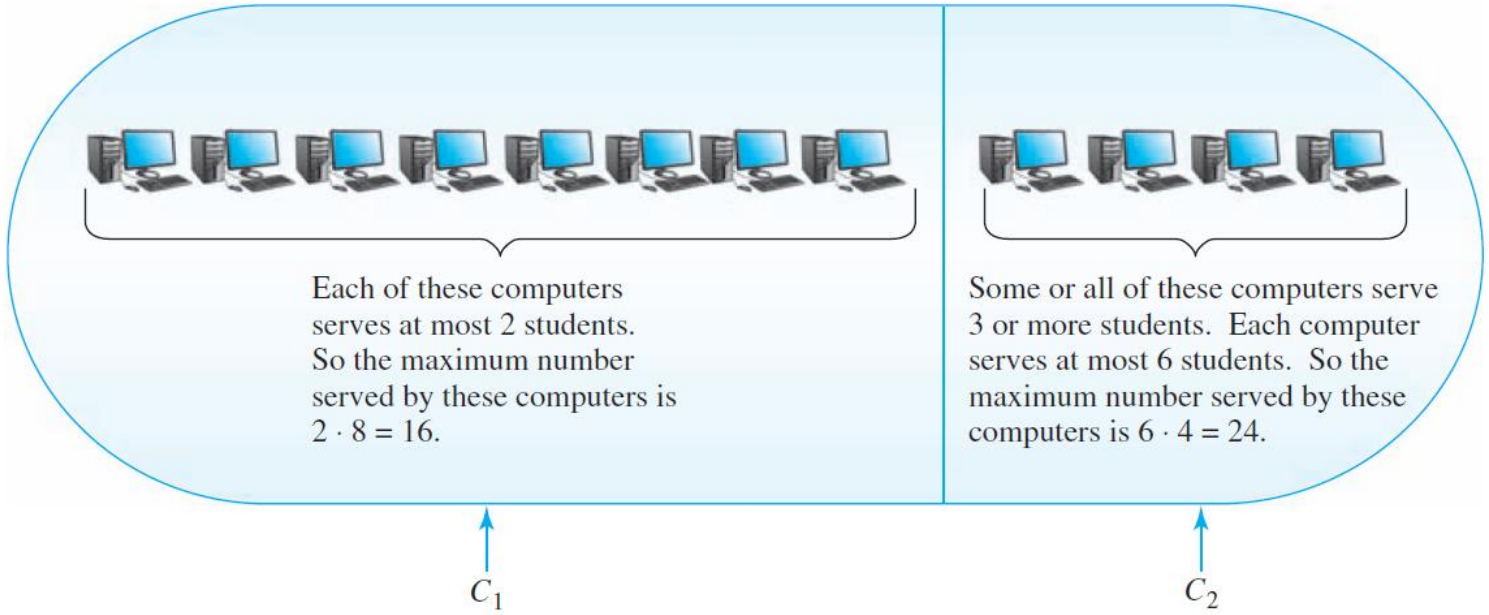


Figure 9.4.3 The set of 12 computers

Since at most 2 students are served by any one computer in C_1, by the generalized pigeonhole principle (contrapositive form), the computers in set C_1 serve at most $2 \times 8 = 16$ students.

Since at most 6 students are served by any one computer in C_2, by the generalized pigeonhole principle (contrapositive form), the computers in set C_2 serve at most $6 \times 4 = 24$ students.

Total number of students served by the computers is at most $24 + 16 = 40$.

This contradicts that there are 42 students, each served by a computer.

Therefore, the supposition is false.

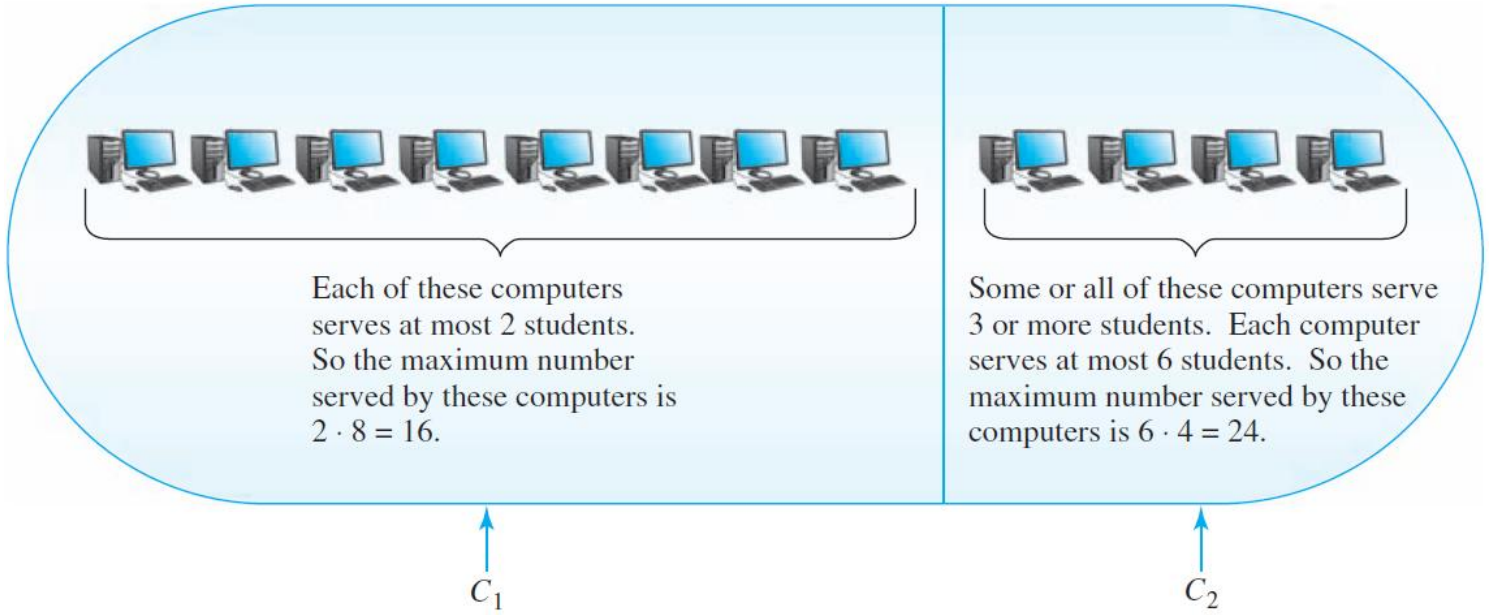


Figure 9.4.3 The set of 12 computers

Generalized Pigeonhole Principle (contrapositive)

Example 12: There are 42 students who are to share 12 computers. Each student uses exactly 1 computer, and no computer is used by more than 6 students. Show that at least 5 computers are used by 3 or more students.

Using a Direct Argument: Let k be the number of computers used by 3 or more students. *[We must show that $k \geq 5$.]*

Because each computer is used by at most 6 students, these computers are used by at most $6k$ students (by the generalized PHP (contrapositive form)).

Each of the remaining $12 - k$ computers is used by at most 2 students.

Taken together, they are used by at most $2(12 - k) = 24 - 2k$ students (again, by the generalized PHP).

Thus the maximum number of students served by the computers is $6k + (24 - 2k) = 4k + 24$.

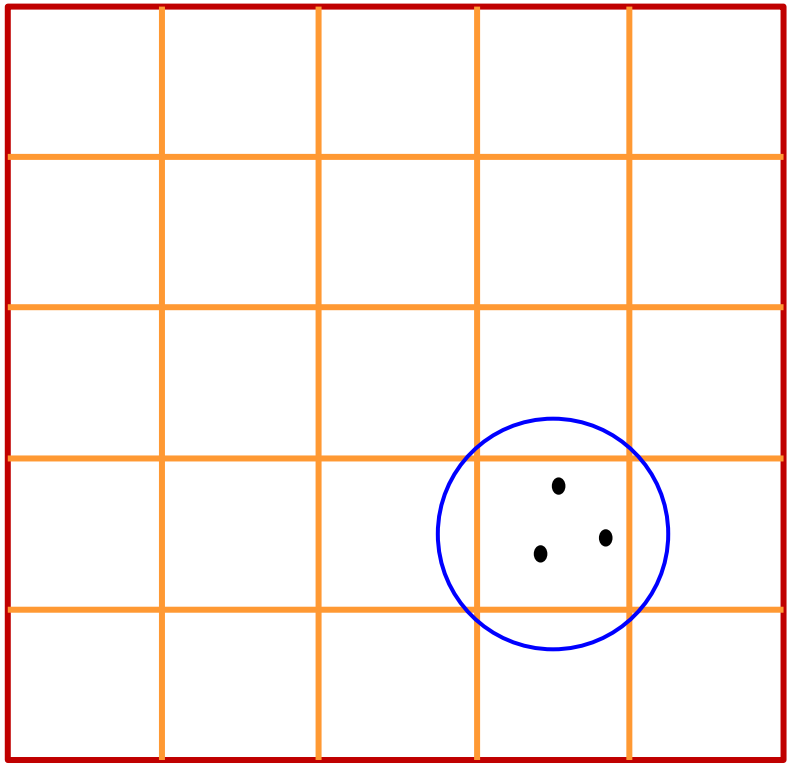
Because 42 students are served by the computers, $4k + 24 \geq 42$.

Solving for k gives $k \geq 4.5$, and since k is an integer, this implies that $k \geq 5$.

Past-Year's Exam Question!

AY2016/17 Sem2

Prove that if you randomly put 51 points inside a unit square, there are always 3 points that can be covered by a circle of radius 1/7.



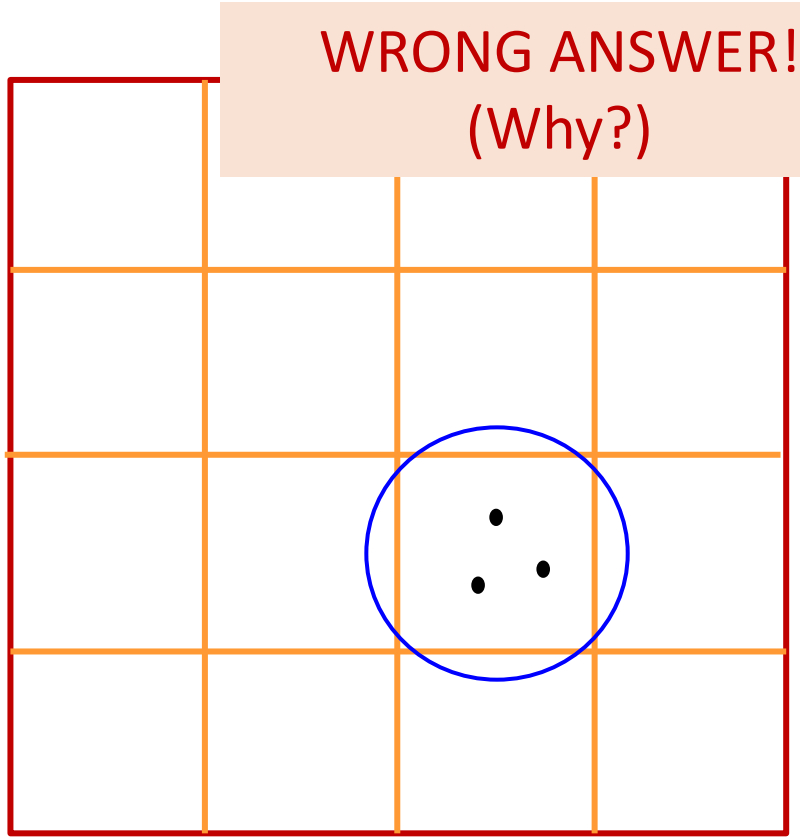
1. Divide the unit square into 25 equal smaller squares of side 1/5 each.
2. Now, at least one of these small squares would contain at least 3 points. (Why?)
3. Now, the circle circumvented around the small square with the 3 points inside also contains these 3 points as it has radius

$$\sqrt{\left(\frac{1}{10}\right)^2+\left(\frac{1}{10}\right)^2}=\sqrt{\frac{1}{50}}<\sqrt{\frac{1}{49}}=\frac{1}{7}$$

Past-Year's Exam Question!

AY2016/17 Sem2

Prove that if you randomly put 51 points inside a unit square, there are always 3 points that can be covered by a circle of radius 1/7.



WRONG ANSWER!
(Why?)

1. Divide the unit square into 16 equal smaller squares of side 1/4 each.
2. Now, at least one of these small squares would contain at least 3 points. (Why?)
3. Now, the area of a small square is 1/16=0.0625, which is smaller than the area of a circle of radius 1/7 which is 0.0641.

Although the area of the small square < area of the circle, the radius of the smallest circle that covers the small square completely is

$$\sqrt{\left(\frac{1}{8}\right)^2+\left(\frac{1}{8}\right)^2}=\sqrt{\frac{1}{32}}>\frac{1}{7}$$

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